



*Great
Astronomers:*

Galileo Galilei

Sir Robert S. Ball



Great Astronomers: Galileo Galilei

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About Ball:

Sir Robert Stawell Ball, Fellow of The Royal Society, (1 July 1840, Dublin – 25 November 1913, Cambridge) was an Irish astronomer. He worked for Lord Rosse from 1865 to 1867. In 1867 he became Professor of Applied Mathematics at the Royal College of Science in Dublin. In 1874 Ball was appointed Royal Astronomer of Ireland and Andrews Professor of Astronomy in the University of Dublin at Dunsink Observatory.[1] In 1892 he was appointed Lowndean Professor of Astronomy and Geometry at Cambridge University at the same time becoming director of the Cambridge Observatory. His lectures, articles, and books (e.g. *Starland* and *The Story of the Heavens*) were mostly popular and simple in style. However, he also published books on mathematical astronomy such as *A Treatise on Spherical Astronomy*. His main interest was mathematics and he devoted much of his spare time to his "Screw theory". He served for a time as President of the Quaternion Society. His work *The Story of the Heavens* is mentioned in the "Lestrygonians" chapter of James Joyce's *Ulysses*. He was the son of naturalist Robert Ball and Amelia Gresley Hellicar. (Biography from Wikipedia: http://en.wikipedia.org/wiki/Robert_Stawell_Ball)

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Chapter 1

AMONG the ranks of the great astronomers it would be difficult to find one whose life presents more interesting features and remarkable vicissitudes than does that of Galileo. We may consider him as the patient investigator and brilliant discoverer. We may consider him in his private relations, especially to his daughter, Sister Maria Celeste, a woman of very remarkable character ; and we have also the pathetic drama at the close of Galileo's life, when the philosopher drew down upon himself the thunders of the Inquisition.

The materials for the sketch of this astonishing man are sufficiently abundant. We make special use in this place of those charming letters which his daughter wrote to him from her convent home. More than a hundred of these have been preserved, and it may well be doubted whether any more beautiful and touching series of letters addressed to a parent by a dearly loved child have ever been written. An admirable account of this correspondence is contained in a little book entitled "The Private Life of Galileo," published anonymously by Messrs. Macmillan in 1870, and I have been much indebted to the author of that volume for many of the facts contained in this chapter.

Galileo was born at Pisa, on 18th February, 1564. He was the eldest son of Vincenzo de Bonajuti de Galilei, a Florentine noble. Notwithstanding his illustrious birth and descent, it would seem that the home in which the great philosopher's childhood was spent was an impoverished one. It was obvious at least that the young Galileo would have to be provided with some profession by which he might earn a livelihood. From his father he derived both by inheritance and by precept a keen taste for music, and it appears that he became an excellent performer on the lute. He was also endowed with considerable artistic power, which he cultivated diligently. Indeed, it would seem that for some time the future astronomer entertained the idea of devoting himself to painting as a profession. His father, however, decided that he should study medicine. Accordingly, we find that when Galileo was seventeen years of age, and had added a knowledge of Greek and Latin to his acquaintance with the fine arts, he was duly entered at the University of Pisa.

Here the young philosopher obtained some inkling of mathematics, whereupon he became so much interested in this branch of science, that he begged to be allowed to study geometry. In compliance with his request, his father permitted a tutor to be engaged for this purpose ; but he did so with reluctance, fearing that the attention of the young student might thus be withdrawn from that medical work which was regarded as his primary occupation. The event speedily proved that these anxieties were not without some justification. The propositions of Euclid proved so engrossing to Galileo that it was thought wise to avoid further distraction by terminating the mathematical tutor's engagement. But it was too late for the desired end to be attained. Galileo had now made such progress that he was able to continue his geometrical studies by himself. Presently he advanced to that famous 47th proposition which won his lively admiration, and on he went until he had mastered the six books of Euclid, which was a considerable achievement for those days.

The diligence and brilliance of the young student at Pisa did not, however, bring him much credit with the University authorities. In those days the doctrines of Aristotle were regarded as the embodiment of all human-wisdom in natural science as well as in everything else. It was regarded as the duty of every student to learn Aristotle off by heart, and any disposition to doubt or even to question the doctrines of the venerated teacher was regarded as intolerable presumption. But young Galileo had the audacity to think for himself about the laws of nature. He would not take any assertion of fact on the authority of Aristotle when he had the means of questioning nature directly as to its truth or falsehood. His teachers thus came to regard him as a somewhat misguided youth, though they could not but respect the unflagging industry with which he amassed all the knowledge he could acquire.

We are so accustomed to the use of pendulums in our clocks, that perhaps we do not often realise that the introduction of this method of regulating time-pieces was really a notable invention worthy the fame of the great astronomer to whom it was due. It appears that sitting one day in the Cathedral of Pisa, Galileo's attention became concentrated on the swinging of a chandelier which hung from the ceiling. It struck him as a significant point, that whether the arc through which the pendulum oscillated was a long one or a short one, the time occupied in each vibration was sensibly the same. This suggested to the thoughtful observer that a pendulum would afford the means by which a time-keeper might be controlled, and accordingly Galileo

constructed for the first time a clock on this principle. The immediate object sought in this apparatus was to provide a means of aiding physicians in counting the pulses of their patients.

The talents of Galileo having at length extorted due recognition from the authorities, he was appointed, at the age of twenty-five, Professor of Mathematics at the University of Pisa. Then came the time when he felt himself strong enough to throw down the gauntlet to the adherents of the old philosophy. As a necessary part of his doctrine on the movement of bodies, Aristotle had asserted that the time occupied by a stone in falling depends upon its weight, so that the heavier the stone the less time would it require to fall from a certain height to the earth. It might have been thought that a statement so easily confuted by the simplest experiments could never have maintained its position in any accepted scheme of philosophy. But Aristotle had said it, and to any one who ventured to express a doubt the ready sneer was forthcoming, "Do you think yourself a cleverer man than Aristotle?"

Galileo determined to demonstrate in the most emphatic manner the absurdity of a doctrine which had for centuries received the sanction of the learned. The summit of the Leaning Tower of Pisa offered a highly dramatic site for the great experiment. The youthful professor let fall from the overhanging top a large heavy body and a small light body simultaneously. According to Aristotle the large body ought to have reached the ground much sooner than the small one, but such was found not to be the case. In the sight of a large concourse of people the simple fact was demonstrated that the two bodies fell side by side, and reached the ground at the same time. Thus the first great step was taken in the overthrow of that preposterous system of unquestioning adhesion to dogma, which had impeded the development of the knowledge of nature for nearly two thousand years.

This revolutionary attitude towards the ancient beliefs was not calculated to render Galileo's relations with the University authorities harmonious. He had also the misfortune to make enemies in other quarters. Don Giovanni de Medici, who was then the Governor of the Port of Leghorn, had designed some contrivance by which he proposed to pump out a dock. But Galileo showed up the absurdity of this enterprise in such an aggressive manner that Don Giovanni took mortal offence, nor was he mollified when the truths of Galileo's criticisms were abundantly verified by the total failure of his ridiculous invention. In various ways Galileo was made to feel his position at Pisa so unpleasant that he was at length compelled to abandon his chair in the

University. The active exertions of his friends, of whom Galileo was so fortunate as to have had throughout his life an abundant supply, then secured his election to the Professorship of Mathematics at Padua, whither he went in 1592. ; It was in this new position that Galileo entered on that marvellous career of investigation which was destined to revolutionize science. The zeal with which he discharged his professorial duties was indeed of the most unremitting character. He speedily drew such crowds to listen to his discourses on Natural Philosophy that his lecture-room was filled to overflowing. He also received many private pupils in his house for special instruction. Every moment that could be spared from these labours was devoted to his private study and to his incessant experiments.

Chapter 2

Like many another philosopher who has greatly extended our knowledge of nature, Galileo had a remarkable aptitude for the invention of instruments designed for philosophical research. To facilitate his practical work, we find that in 1599 he had engaged a skilled workman who was to live in his house, and thus be constantly at hand to try the devices for ever springing from Galileo's fertile brain. Among the earliest of his inventions appears to have been the thermometer, which he constructed in 1602. No doubt this apparatus in its primitive form differed in some respects from the contrivance we call by the same name. Galileo at first employed water as the agent, by the expansion of which the temperature was to be measured. He afterwards saw the advantage of using spirits for the same purpose. It was not until about half a century later that mercury came to be recognised as the liquid most generally suitable for the thermometer.

The time was now approaching when Galileo was to make that mighty step in the advancement of human knowledge which followed on the application of the telescope to astronomy. As to how his idea of such an instrument originated, we had best let him tell us in his own words. The passage is given in a letter which he writes to his brother-in-law, Landucci.

" I write now because I have a piece of news for you, though whether you will be glad or sorry to hear it I cannot say; for I have now no hope of returning to my own country, though the occurrence which has destroyed that hope has had results both useful and honourable. You must know, then, that two months ago there was a report spread here that in Flanders some one had presented to Count Maurice of Nassau a glass manufactured in such a way as to make distant objects appear very near, so that a man at the distance of two miles could be clearly seen. This seemed to me so marvellous that I began to think about it. As it appeared to me to have a foundation in the Theory of Perspective, I set about contriving how to make it, and at length I found out, and have

succeeded so well that the one I have made is far superior to the Dutch telescope. It was reported in Venice that I had made one, and a week since I was commanded to show it to his Serenity and to all the members of the senate, to their infinite amazement. Many gentlemen and senators, even the oldest, have ascended at various times the highest bell-towers in Venice to spy out ships at sea making sail for the mouth of the harbour, and have seen them clearly, though without my telescope they would have been invisible for more than two hours. The effect of this instrument is to show an object at a distance of say fifty miles, as if it were but five miles."

The remarkable properties of the telescope at once commanded universal attention among intellectual men. Galileo received applications from several quarters for his new instrument, of which it would seem that he manufactured a large number to be distributed as gifts to various illustrious personages.

But it was reserved for Galileo himself to make that application of the instrument to the celestial bodies by which its peculiar powers were to inaugurate the new era in astronomy. The first discovery that was made in this direction appears to have been connected with the number of the stars. Galileo saw to his amazement that through his little tube he could count ten times as many stars in the sky as his unaided eye could detect. Here was, indeed, a surprise. We are now so familiar with the elementary facts of astronomy that it is not always easy to realise how the heavens were interpreted by the observers in those ages prior to the invention of the telescope. We can hardly, indeed, suppose that Galileo, like the majority of those who ever thought of such matters, entertained the erroneous belief that the stars were on the surface of a sphere at equal distances from the observer. No one would be likely to have retained his belief in such a doctrine when he saw how the number of visible stars could be increased tenfold by means of Galileo's telescope. It would have been almost impossible to refuse to draw the inference that the stars thus brought into view were still more remote objects which the telescope was able to reveal, just in the same way as it showed certain ships to the astonished Venetians, when at the time these ships were beyond the reach of unaided vision.

Galileo's celestial discoveries now succeeded each other rapidly. That beautiful Milky Way, which has for ages been the object of admiration to all lovers of nature, never disclosed its true nature to the eye of man till the

astronomer of Padua turned on it his magic tube. The splendid zone of silvery light was then displayed as star-dust scattered over the black background of the sky. It was observed that though the individual stars were too small to be seen severally without optical aid, yet such was their incredible number that the celestial radiance produced that luminosity with which every star-gazer was so familiar.

But the greatest discovery made by the telescope in these early days, perhaps, indeed, the greatest discovery that the telescope has ever accomplished, was the detection of the system of four satellites revolving around the great planet Jupiter. This phenomenon was so wholly unexpected by Galileo that, at first, he could hardly believe his eyes. However, the reality of the existence of a system of four moons attending the great planet was soon established beyond all question. Numbers of great personages crowded to Galileo to see for themselves this beautiful miniature representing the sun with its system of revolving planets.

Of course there were, as usual, a few incredulous people who refused to believe the assertion that four more moving bodies had to be added to the planetary system. They scoffed at the notion; they said the satellites may have been in the telescope, but that they were not in the sky. One sceptical philosopher is reported to have affirmed, that even if he saw the moons of Jupiter himself he would not believe in them, as their existence was contrary to the principles of common-sense !

There can be no doubt that a special significance attached to the new discovery at this particular epoch in the history of science. It must be remembered that in those days the doctrine of Copernicus, declaring that the sun, and not the earth, was the centre of the system, that the earth revolved on its axis once a day, and that it described a mighty circle round the sun once a year, had only recently been promulgated. This new view of the scheme of nature had been encountered with the most furious opposition. It may possibly have been that Galileo himself had not felt quite confident in the soundness of the Copernican theory, prior to the discovery of the satellites of Jupiter. But when a picture was there exhibited in which a number of relatively small globes were shown to be revolving around a single large globe in the centre, it seemed impossible not to feel that the beautiful spectacle so displayed was an emblem of the relations of the planets to the sun. It was thus made manifest to Galileo that the Copernican theory of the planetary system must be the true one. The momentous import of this opinion

upon the future welfare of the great philosopher will presently appear.

Chapter 3

It would seem that Galileo regarded his residence at Padua as a state of undesirable exile from his beloved Tuscany. He had always a yearning to go back to his own country, and at last the desired opportunity presented itself. For now that Galileo's fame had become so great, the Grand Duke of Tuscany desired to have the philosopher resident at Florence, in the belief that he would shed lustre on the Duke's dominions. Overtures were accordingly made to Galileo, and the consequence was that in 1616 we find him residing at Florence, bearing the title of Mathematician and Philosopher to the Grand Duke.

Two daughters, Polissena and Virginia, and one son, Vincenzo, had been born to Galileo in Padua. It was the custom in those days that as soon as the daughter of an Italian gentleman had grown up, her future career was somewhat summarily decided. Either a husband was to be forthwith sought out, or she was to enter the convent with the object of taking the veil as a professed nun. It was arranged that the two daughters of Galileo, while still scarcely more than children, should both enter the Franciscan convent of St. Matthew, at Arcetri. The elder daughter, Polissena, took the name of Sister Maria Celeste, while Virginia became Sister Arcangela. The latter seems to have been always delicate and subject to prolonged melancholy, and she is of but little account in the narrative of the life of Galileo. But Sister Maria Celeste, though never leaving the convent, managed to preserve a close intimacy with her beloved father. This was maintained only partly by Galileo's visits, which were very irregular and were, indeed, often suspended for long intervals. But his letters to this daughter were evidently frequent and affectionate, especially in the latter part of his life. Most unfortunately, however, all his letters have been lost. There are grounds for believing that they were deliberately destroyed when Galileo was seized by the Inquisition, lest they should have been used as evidence against him, or lest they should have compromised the convent where they were received. But Sister Maria Celeste's letters to her father have happily been preserved, and most touching

these letters are. We can hardly read them without thinking how the sweet and gentle nun would have shrunk from the idea of their publication.

Her loving little notes to her " dearest lord and father," as she used affectionately to call Galileo, were almost invariably accompanied by some gift, trifling it may be, but always the best the poor nun had to bestow. The tender grace of these endearing communications was all the more precious to him from the fact that the rest of Galileo's relatives were of quite a worthless description. He always acknowledged the ties of his kindred in the most generous way, but their follies and their vices, their selfishness and their importunities, were an incessant source of annoyance to him, almost to the last day of his life.

On 19th December, 1625, Sister Maria Celeste writes:—

" I send two baked pears for these days of vigil. But as the greatest treat of all, I send you a rose, which ought to please you extremely, seeing what a rarity it is at this season; and with the rose you must accept its thorns, which represent the bitter passion of our Lord, whilst the green leaves represent the hope we may entertain that through the same sacred passion we, having passed through the darkness of the short winter of our mortal life, may attain to the brightness and felicity of an eternal spring in heaven."

When the wife and children of Galileo's shiftless brother came to take up their abode in the philosopher's home, Sister Maria Celeste feels glad to think that her father has now some one who, however imperfectly, may fulfil the duty of looking after him. A graceful note on Christmas Eve accompanies her little gifts. She hopes that—

" In these holy days the peace of God may rest on him and all the house. The largest collar and sleeves I mean for Albertino, the other two for the two younger boys, the little dog for baby, and the cakes for everybody, except the spice-cakes, which are for you. Accept the good-will which would readily do much more."

The extraordinary forbearance with which Galileo continually placed his time, his purse, and his influence at the service of those who had repeatedly proved themselves utterly unworthy of his countenance, is thus commented on by the good nun:—

" Now it seems to me, dearest lord and father, that your lordship is walking in the right path, since you take hold of every occasion that presents itself to shower continual benefits on those who only repay you with ingratitude. This is an action which is all the more virtuous and perfect as it is the more difficult."

When the plague was raging in the neighbourhood, the loving daughter's solicitude is thus shown :—

"I send you two pots of electuary^[1] as a preventive against the plague. The one without the label consists of dried figs, walnuts, rue, and salt, mixed together with honey. A piece of the size of a walnut to be taken in the morning, fasting, with a little Greek wine."

The plague increasing still more, Sister Maria Celeste obtained, with much difficulty, a small quantity of a renowned liqueur, made by Abbess Ursula, an exceptionally saintly nun. This she sends to her father with the words :—

"I pray your lordship to have faith in this remedy. For if you have so much faith in my poor miserable prayers, much more may you have in those of such a holy person ; indeed, through her merits you may feel sure of escaping all danger from the plague."

Whether Galileo took the remedy we do not know, but at all events he escaped the plague.

From Galileo's new home in Florence the telescope was again directed to the skies, and again did astounding discoveries reward the astronomer's labours. The great success which he had met with in studying Jupiter naturally led Galileo to look at Saturn. Here he saw a spectacle which was sufficiently amazing, though he failed to interpret it accurately. It was quite manifest that Saturn did not exhibit a simple circular disc like Jupiter, or like Mars. It seemed to Galileo as if the planet consisted of three bodies, a large globe in the centre, and a smaller one on each side. The enigmatical nature of the discovery led Galileo to announce it in an enigmatical manner. He published a string of letters which, when duly transposed, made up a sentence which affirmed that the planet Saturn was threefold. Of course we now know that this remarkable appearance of the planet was due to the two projecting portions of the ring. With the feeble power of Galileo's telescope, these

seemed merely like small globes or appendages to the large central body.

The last of Galileo's great astronomical discoveries related to the libration of the moon. I think that the detection of this phenomenon shows his acuteness of observation more remarkably than does any one of his other achievements with the telescope. It is well known that the moon constantly keeps the same face turned towards the earth. When, however, careful measurements have been made with regard to the spots and marks on the lunar surface, it is found that there is a slight periodic variation which permits us to see now a little to the east or to the west, now a little to the north or to the south of the average lunar disc.

Chapter 4

But the circumstances which make the career of Galileo so especially interesting from the biographer's point of view, are hardly so much the triumphs that he won as the sufferings that he endured. The sufferings and the triumphs were, however, closely connected, and it is fitting that we should give due consideration to what was perhaps the greatest drama in the history of science.

On the appearance of the immortal work of Copernicus, in which it was taught that the earth rotated on its axis, and that the earth, like the other planets, revolved round the sun, orthodoxy stood aghast. The Holy Roman Church submitted this treatise, which bore the name "*De Revolutionibus Orbium Coelestium*," to the Congregation of the Index.^[2] After due examination it was condemned as heretical in 1615. Galileo was suspected, on no doubt excellent grounds, of entertaining the objectionable views of Copernicus. He was accordingly privately summoned before Cardinal Bellarmine on 26th February, 1616, and duly admonished that he was on no account to teach or to defend the obnoxious doctrines. Galileo was much distressed by this intimation. He felt it a serious matter to be deprived of the privilege of discoursing with his friends about the Copernican system, and of instructing his disciples in the principles of the great theory of whose truth he was perfectly convinced. It pained him, however, still more to think, devout Catholic as he was, that such suspicions of his fervent allegiance to his Church should ever have existed, as were implied by the words and monitions of Cardinal Bellarmine.

In 1616, Galileo had an interview with Pope Paul V., who received the great astronomer very graciously, and walked up and down with him in conversation for three-quarters of an hour. Galileo complained to his Holiness of the attempts made by his enemies to embarrass him with the authorities of the Church, but the Pope bade him be comforted. His Holiness had himself no doubts of Galileo's orthodoxy, and he assured him that the Congregation of the Index should give Galileo no further trouble so long as

Paul V. was in the chair of St. Peter.

On the death of Paul V. in 1623, Maffeo Barberini was elected Pope, as Urban VIII. This new Pope, while a cardinal, had been an intimate friend of Galileo's, and had indeed written Latin verses in praise of the great astronomer and his discoveries. It was therefore not unnatural for Galileo to think that the time had arrived when, with the use of due circumspection, he might continue his studies and his writings, without fear of incurring the displeasure of the Church. Indeed, in 1624, one of Galileo's friends writing from Rome, urges Galileo to visit the city again, and added that—

"Under the auspices of this most excellent, learned, and benignant Pontiff, science must flourish. Your arrival will be welcome to his Holiness. He asked me if you were coming, and when, and in short, he seems to love and esteem you more than ever."

The visit was duly paid, and when Galileo returned to Florence, the Pope wrote a letter from which the following is an extract, commending the philosopher to the good offices of the young Ferdinand, who had shortly before succeeded his father in the Grand Duchy of Tuscany.

" We find in Galileo not only literary distinction, but also the love of piety, and he is also strong in those qualities by which the pontifical good-will is easily obtained. And now, when he has been brought to this city to congratulate us on our elevation, we have very lovingly embraced him ; nor can we suffer him to return to the country whither your liberality calls him, without an ample provision of pontifical love. And that you may know how dear he is to us, we have willed to give him this honourable testimonial of virtue and piety. And we further signify that every benefit which you shall confer upon him, imitating or even surpassing your father's liberality, will conduce to our gratification."

The favourable reception which had been accorded to him by Pope Urban VIII. seems to have led Galileo to expect that there might be some corresponding change in the attitude of the Papal authorities on the great question of the stability of the earth. He accordingly proceeded with the preparation of the chief work of his life, "The Dialogue of the Two Systems." It was submitted for inspection by the constituted authorities. The Pope himself thought that, if a few conditions which he laid down were duly

complied with, there could be no objection to the publication of the work. In the first place, the title of the book was to be so carefully worded as to show plainly that the Copernican doctrine was merely to be regarded as an hypothesis, and not as a scientific fact. Galileo was also instructed to conclude the book with special arguments which had been supplied by the Pope himself, and which appeared to his Holiness to be quite conclusive against the new doctrine of Copernicus.

Formal leave for the publication of the Dialogue was then given to Galileo by the Inquisitor General, and it was accordingly sent to the press. It might be thought that the anxieties of the astronomer about his book would then have terminated. As a matter of fact, they had not yet seriously begun. Riccardi, the Master of the Sacred Palace, having suddenly had some further misgivings, sent to Galileo for the manuscript while the work was at the printer's, in order that the doctrine it implied might be once again examined. Apparently, Riccardi had come to the conclusion that he had not given the matter sufficient attention, when the authority to go to press had been first and, perhaps, hastily given. Considerable delay in the issue of the book was the result of these further deliberations. At last, however, in June, 1632, Galileo's great work, "The Dialogue of the Two Systems," was produced for the instruction of the world, though the occasion was fraught with ruin to the immortal author.

The book, on its publication, was received and read with the greatest avidity. But presently the Master of the Sacred Palace found reason to regret that he had given his consent to its appearance. He accordingly issued a peremptory order to sequester every copy in Italy. This sudden change in the Papal attitude towards Galileo formed the subject of a strong remonstrance addressed to the Roman authorities by the Grand Duke of Tuscany. The Pope himself seemed to have become impressed all at once with the belief that the work contained matter of an heretical description. The general interpretation put upon the book seems to have shown the authorities that they had mistaken its true tendency, notwithstanding the fact that it had been examined again and again by theologians deputed for the duty. To the communication from the Grand Duke the Pope returned answer, that he had decided to submit the book to a congregation of "learned, grave, and saintly men," who would weigh every word in it. The views of his Holiness personally on the subject were expressed in his belief that the Dialogue contained the most perverse matter that could come into a reader's hands.

The Master of the Sacred Palace was greatly blamed by the authorities for having given his sanction to its issue. He pleaded that the book had not been printed in the precise terms of the original manuscript which had been submitted to him. It was also alleged that Galileo had not adhered to his promise of inserting properly the arguments which the Pope himself had given in support of the old and orthodox view. One of these had, no doubt, been introduced, but, so far from mending Galileo's case, it had made matters really look worse for the poor philosopher. The Pope's argument had been put into the mouth of one of the characters in the Dialogue named "Simplicio." Galileo's enemies maintained that by adopting such a method for the expression of his Holiness's opinion, Galileo had intended to hold the Pope himself up to ridicule. Galileo's friends maintained that nothing could have been farther from his intention. It seems, however, highly probable that the suspicions thus aroused had something to say to the sudden change of front on the part of the Papal authorities.

On 1st October, 1632, Galileo received an order to appear before the Inquisition at Rome on the grave charge of heresy. Galileo, of course, expressed his submission, but pleaded for a respite from compliance with the summons, on the ground of his advanced age and his failing health. The Pope was, however, inexorable; he said that he had warned Galileo of his danger while he was still his friend. The command could not be disobeyed. Galileo might perform the journey as slowly as he pleased, but it was imperatively necessary for him to set forth, and at once.

On 20th January, 1633, Galileo started on his weary journey to Rome, in compliance with this peremptory summons. On 13th February he was received as the guest of Niccolini, the Tuscan ambassador, who had acted as his wise and ever-kind friend throughout the whole affair. It seemed plain that the Holy Office were inclined to treat Galileo with as much clemency and consideration as was consistent with the determination that the case against him should be proceeded with to the end. The Pope intimated that in consequence of his respect for the Grand Duke of Tuscany he should permit Galileo to enjoy the privilege, quite unprecedented for a prisoner charged with heresy, of remaining as an inmate in the ambassador's house. He ought, strictly, to have been placed in the dungeons of the Inquisition. When the examination of the accused had actually commenced, Galileo was confined, not, indeed, in the dungeons, but in comfortable rooms at the Holy Office.

By the judicious and conciliatory language of submission which Niccolini

had urged Galileo to use before the Inquisitors, they were so far satisfied that they interceded with the Pope for his release. During the remainder of the trial Galileo was accordingly permitted to go back to the ambassador's, where he was most heartily welcomed. Sister Maria Celeste, evidently thinking this meant that the whole case was at an end, thus expresses herself:—

" The joy that your last dear letter brought me, and the having to read it over and over to the nuns, who made quite a jubilee on hearing its contents, put me into such an excited state that at last I got a severe attack of headache."

In his defence Galileo urged that he had already been acquitted in 1616 by Cardinal Bellarmine, when a charge of heresy was brought against him, and he contended that anything he might now have done, was no more than he had done on the preceding occasion, when the orthodoxy of his doctrines received solemn confirmation. The Inquisition seemed certainly inclined to clemency, but the Pope was not satisfied. Galileo was accordingly summoned again on the 21st June. He was to be threatened with torture if he did not forthwith give satisfactory explanations as to the reasons which led him to write the Dialogue. In this proceeding the Pope assured the Tuscan ambassador that he was treating Galileo with the utmost consideration possible in consequence of his esteem and regard for the Grand Duke, whose servant Galileo was. It was, however, necessary that some exemplary punishment be meted out to the astronomer, inasmuch as by the publication of the Dialogue he had distinctly disobeyed the injunction of silence laid upon him by the decree of 1616. Nor was it admissible for Galileo to plead that his book had been sanctioned by the Master of the Sacred College, to whose inspection it had been again and again submitted. It was held, that if the Master of the Sacred College had been unaware of the solemn warning the philosopher had already received sixteen years previously, it was the duty of Galileo to have drawn his attention to that fact.

On the 22nd June, 1633, Galileo was led to the great hall of the Inquisition, and compelled to kneel before the cardinals there assembled and hear his sentence. In a long document, most elaborately drawn up, it is definitely charged against Galileo that, in publishing the Dialogue, he committed the essentially grave error of treating the doctrine of the earth's motion as open to discussion. Galileo knew, so the document affirmed, that the Church had

emphatically pronounced this notion to be contrary to Holy Writ, and that for him to consider a doctrine so stigmatized as having any shadow of probability in its favour was an act of disrespect to the authority of the Church which could not be overlooked. It was also charged against Galileo that in his Dialogue he has put the strongest arguments into the mouth, not of those who supported the orthodox doctrine, but of those who held the theory as to the earth's motion which the Church had so deliberately condemned.

After due consideration of the defence made by the prisoner, it was thereupon decreed that he had rendered himself vehemently suspected of heresy by the Holy Office, and in consequence had incurred all the censures and penalties of the sacred canons, and other decrees promulgated against such persons. The graver portion of these punishments would be remitted, if Galileo would solemnly repudiate the heresies referred to by an abjuration to be pronounced by him in the terms laid down.

At the same time it was necessary to mark, in some emphatic manner, the serious offence which had been committed, so that it might serve both as a punishment to Galileo and as a warning to others. It was accordingly decreed that he should be condemned to imprisonment in the Holy Office during the pleasure of the Papal authorities, and that he should recite once a week for three years the seven Penitential Psalms.

Then followed that ever-memorable scene in the great hall of the Inquisition, in which the aged and infirm Galileo, the inventor of the telescope and the famous astronomer, knelt down to abjure before the most eminent and reverend Lords Cardinal, Inquisitors General throughout the Christian Republic against heretical depravity. With his hands on the Gospels, Galileo was made to curse and detest the false opinion that the sun was the centre of the universe and immovable, and that the earth was not the centre of the same, and that it moved. He swore that for the future he will never say nor write such things as may bring him under suspicion, and that if he does so he submits to all the pains and penalties of the sacred canons. This abjuration was subsequently read in Florence before Galileo's disciples, who had been specially summoned to attend.

It has been noted that neither on the first occasion, in 1616, nor on the second in 1633, did the reigning Pope sign the decrees concerning Galileo. The contention has accordingly been made that Paul V. and Urban VIII. are both alike vindicated from any technical responsibility for the attitude of the Romish Church towards the Copernican doctrines. The significance of this

circumstance has been commented on in connection with the doctrine of the infallibility of the Pope.

We can judge of the anxiety felt by Sister Maria Celeste about her beloved father during these terrible trials. The wife of the ambassador Niccolini, Galileo's steadfast friend, most kindly wrote to give the nun whatever quieting assurances the case would permit. There is a renewed flow of these touching epistles from the daughter to her father. Thus she sends word—

" The news of your fresh trouble has pierced my soul with grief all the more that it came quite unexpectedly."

And again, on hearing that he had been permitted to leave Rome, she writes—

"I wish I could describe the rejoicing of all the mothers and sisters on hearing of your happy arrival at Siena. It was indeed most extraordinary. On hearing the news the Mother Abbess and many of the nuns ran to me, embracing me and weeping for joy and tenderness."

The sentence of imprisonment was at first interpreted leniently by the Pope. Galileo was allowed to reside in qualified duration in the archbishop's house at Siena. Evidently the greatest pain that he endured arose from the forced separation from that daughter, whom he had at last learned to love with an affection almost comparable with that she bore to him. She had often told him that she never had any pleasure equal to that with which she rendered any service to her father. To her joy, she discovers that she can relieve him from the task of reciting the seven Penitential Psalms which had been imposed as a penance:—

" I began to do this a while ago," she writes, " and it gives me much pleasure. First, because I am persuaded that prayer in obedience to Holy Church must be efficacious ; secondly, in order to save you the trouble of remembering it. If I had been able to do more, most willingly would I have entered a straiter prison than the one I live in now, if by so doing I could have set you at liberty."

Sister Maria Celeste was gradually failing in health, but the great privilege was accorded to her of being able once again to embrace her beloved lord and

master. Galileo had, in fact, been permitted to return to his old home; but on the very day when he heard of his daughter's death came the final decree directing him to remain in his own house in perpetual solitude.

Amid the advancing infirmities of age, the isolation from friends, and the loss of his daughter, Galileo once again sought consolation in hard work. He commenced his famous dialogue on Motion. Gradually, however, his sight began to fail, and blindness was at last added to his other troubles. On January 2nd, 1638, he writes to Dioduti:—

" Alas, your dear friend and servant, Galileo, has been for the last month perfectly blind, so that this heaven, this earth, this universe, which I by my marvellous discoveries and clear demonstrations have enlarged a hundred thousand times beyond the belief of the wise men of bygone ages, henceforward is for me shrunk into such a small space as is filled by my own bodily sensations."

But the end was approaching—the great philosopher, was attacked by low fever, from which he died on the 8th January, 1643.

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[1] Editor's note: An electuary is a medicinal lozenge.

[2] Editor's note: The Index refers to the list of books forbidden by the Catholic Church. With the Council of Trent (1545-1563), the Church instituted a permanent institution to deal with this subject, although similar lists had been by various Popes and Councils since year 405. (Reference: The Galileo Project <http://galileo.rice.edu/chr/congregation.html> .)



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